22MAT220- Mathematics for Computing-3

Smooth and Spike Signal Seperations using Over Complete Dictionary and ADMM

Group C-3

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DCT Function

Discrete Cosine Transform is a mathematical tool to express a signal in terms of sum of cosines

The Discrete Cosine Transform (DCT) takes a finite-length signal and expresses it as a weighted sum of cosine waves with different frequencies. Given a signal x, of length N, we construct an N×N DCT matrix whose k-th row represents a cosine function of frequency k, scaled by a normalization factor α_k to maintain orthogonality. The element at position (k,n) is given by

$A_{k,n} = \alpha_k \cos[(\pi/N) (n + 1/2) k]$, where n is the sample index and k is the frequency index.

Multiplying this matrix by the signal vector x produces the DCT coefficients X[k], where each coefficient represents the contribution of its corresponding cosine wave to the original signal. Low-frequency coefficients capture the slowly varying trends in the signal, while high-frequency coefficients capture rapid changes, making the DCT particularly useful for compression, where high-frequency components can be reduced or removed with minimal perceptual loss.

$$A_{k,n} = \alpha_k \cos\left[\frac{\pi}{N}\left(n+\frac{1}{2}\right)k\right], \quad \alpha_k = \begin{cases} \sqrt{\frac{1}{N}}, & k = 0, \\ \sqrt{\frac{2}{N}}, & k > 0 \end{cases}$$

```
function A=dct_matrix(N)
    A=zeros(N,N);
```

```
    for k=0:N-1
        for n=0:N-1
            A(k+1,n+1)=cos((pi/N)*(n+0.5)*k);
        end
        if k==0
            alpha=sqrt(1/N);
        else
            alpha=sqrt(2/N);
        end
        A(k+1,:)=A(k+1,:)*alpha;
    end
end
dct_matrix(4)
```

```
ans = 4×4
    0.5000    0.5000    0.5000    0.5000
    0.6533    0.2706   -0.2706   -0.6533
    0.5000   -0.5000   -0.5000    0.5000
    0.2706   -0.6533    0.6533   -0.2706
```

Making an over complete dictionary function

Now we will implement a function to return the basis matrix(over complete dictionary) for a signal as input

According to the paper,

A : Over Complete dicitonary

A=[soft,noise]

where,

soft=dct basis matrix for the pure signal

noise=identity basis matrix for the noise

```
clearvars
function A=ocd(x)
    x=x(:);
    len=length(x);
    soft=dct_matrix(len);
    noise=eye(len);
    A=[soft,noise];
end
ocd([1;2;3;4])
```

```
ans = 4×8
    0.5000    0.5000    0.5000    0.5000    1.0000         0         0         0
    0.6533    0.2706   -0.2706   -0.6533         0    1.0000         0         0
    0.5000   -0.5000   -0.5000    0.5000         0         0    1.0000         0
    0.2706   -0.6533    0.6533   -0.2706         0         0         0    1.0000
```

Objective function and update equations

We have the basis of the signal, we need coefficients for the basis matrix such that the basis when multiplied with coefficient vector, gives out the original signal

$A \longrightarrow$ **Over complete Dictionary(Basis Matrix)**

$x \longrightarrow$ **Original Signal**

$\alpha \longrightarrow$ **Coefficients**

So, $\mathrm{A}\alpha = x$

$\mathrm{A} \in R^{\mathrm{Nx2N}}$

$x \in R^N \qquad \text{where } N \longrightarrow \text{Length of the signal}$

$\alpha \in R^{2N}$

As A is not square matrix, it is definitely not having a Unique Solution.

We need to find a solution that is sparse.

Let objective be,

$\min_\alpha \ ||\alpha||_0$

$\text{such that } A\alpha = x$

But this is not Computationally Tracable, so we take the objective to be L1 norm.

So Let the objective be,

$\min_\alpha \ ||\alpha||_1$

$\text{such that } A\alpha = x$

$\alpha^* = \text{argmin}_\alpha \, ||\alpha||_1, \text{ Where, } \alpha^* \rightarrow \text{Sparse Solution}$

For ADMM, we split into 2 functions,

$$\min_{\alpha,z} \ ||\alpha||_1 + g(z) \quad \text{s.t. } \alpha = z,$$

where

$$g(z) = \begin{cases} 0, & \text{if } Az = x, \\ +\infty, & \text{otherwise.} \end{cases}$$

$g(z) \rightarrow \text{Indicator Function}$

Lagrangian Function of the objective is,

$$L(\alpha, z, \lambda) = ||\alpha||_1 + g(z) + \lambda^T(\alpha - z)$$

Augmented Lagrangian Function is,

$$L_\rho(\alpha, z, \lambda) = ||\alpha||_1 + g(z) + \frac{\rho}{2}||\alpha - z||^2 + \lambda^T(\alpha - z)$$

Update Equations are,

$$\alpha^{k+1} = \text{argmin}_\alpha \, ||\alpha||_1 + \frac{\rho}{2}||\alpha - \left(z^k - \frac{\lambda^k}{\rho}\right)||^2{}_2$$

$$z^{k+1} = \text{argmin}_z \, g(z) + \frac{\rho}{2}||z - \left(\alpha^{k+1} + \frac{1}{\rho}\lambda^k\right)||^2{}_2$$

Let, $y^k = z^k - \frac{\lambda^k}{\rho}$

α update :

There are 3 cases:

1. $\alpha_i > 0, \text{then } |\alpha_i| = \alpha_i$:

then, $f(\alpha_i) = \alpha_i + \frac{\rho}{2}(\alpha_i - y_i)^2$

Differentiate $f(\alpha_i)$ and equate it to 0.

$\implies \alpha_i = y_i - \frac{1}{\rho}$

Only valid if, $y_i > \frac{1}{\rho}$

2. $\alpha_i < 0, \text{then } |\alpha_i| = -\alpha_i$

then, $f(\alpha_i) = -\alpha_i + \frac{\rho}{2}(\alpha_i - y_i)^2$

Differentiate $f(\alpha_i)$ and equate it to 0.

$\implies \alpha_i = y_i + \frac{1}{\rho}$

Only valid, if $y_i < -\frac{1}{\rho}$

3.if $|y_i| \leq \frac{1}{\rho}$ $\text{then}, \alpha_i = 0$

These 3 cases combine to:

$$\alpha_i^{k+1} = \begin{cases} y_i - \frac{1}{p}, & y_i > \frac{1}{\rho}, \\ 0, & |y_i| \leq \frac{1}{\rho}, \\ y_i + \frac{1}{\rho}, & y_i < -\frac{1}{\rho}. \end{cases}$$

This is same as the shrinkage function.

$$\mathcal{S}_\lambda(x) = \begin{cases} x - \lambda, & x > \lambda \\ 0, & |x| \le \lambda \\ x + \lambda, & x < -\lambda \end{cases}$$

$$\mathcal{S}_\lambda(x) = \text{sign}(x) \cdot \max(|x| - \lambda,\ 0)$$

So,

$$\alpha^{k+1} = \mathcal{S}_{\frac{1}{\rho}}\left(z^k - \frac{\lambda^k}{\rho}\right),$$

$$\mathcal{S}_\theta(y)_i = sign(y_i)\,.\max\left(|y_i| - \theta, 0\right).$$

The shrinkage function is the **proximal operator** for the L1 norm. It "shrinks" each coefficient towards zero by subtracting a threshold. If the coefficient's magnitude is smaller than that threshold, it is set to **exactly zero**. If it is larger, its magnitude is reduced by threshold but the sign is kept.

z update :

$$z^{k+1} = \text{argmin}_z \ \ g(z) + \frac{\rho}{2}||\alpha^{k+1} - z\,||_2^2 + \lambda^T\left(\alpha^{k+1} - z\ \right)$$

As, we want Az=x convergence, we can take the value of the indicator function as 0

So,

$$z^{k+1} = \text{argmin}_z \ \ + \frac{\rho}{2}||z - \left(\alpha^{k+1} + \frac{1}{\rho}\lambda^k\right)||_2^2$$

Z can have infinite solutions. But we need to find z which is nearer to $\left(\alpha^{k+1} + \frac{1}{\rho}\lambda^k\right)$. So, this term will be perpendicular to the space of z. So we can project this term onto the space of z.

Now our task is to find the z which is nearer to the term. We can do this by lagrangian multiplier method

Let, $\alpha^{k+1} + \frac{1}{\rho}\lambda^k = d$

Let, the lagrangian equation be,

$$L(z, v) = \frac{1}{2}||z - d||_2^2 + v^T(\text{Az} - x)$$

Differentiating and equating to 0, we get

$z = d - A^t v$ -----equation 1

substitute equation 1 in Az=x

We get,

$$v = (\text{AA}^T)^{-1}(\text{Ad} - x)$$

$$z = d - A^T(\text{AA}^T)^{-1}(\text{Ad} - x)$$

In Final,

$$z^{k+1} = (I - A^{+}A)\left(\alpha^{k+!} + \frac{1}{\rho}\lambda^{k}\right) + A^{+}x$$

Update Equations

$$\alpha^{k+1} = S_{\frac{1}{\rho}}\left(z^{k} - \frac{1}{\rho}\lambda^{k}\right)$$

$$z^{k+1} = (I - A^{+}A)\left(\alpha^{k+!} + \frac{1}{\rho}\lambda^{k}\right) + A^{+}x$$

$$\lambda^{k+1} = \lambda^{k} + \rho(\alpha^{k+!} - z^{k+1})$$

Shrinkage function

$$\mathcal{S}_{\lambda}(x) = \begin{cases} x - \lambda, & x > \lambda \\ 0, & |x| \leq \lambda \\ x + \lambda, & x < -\lambda \end{cases}$$

$$\mathcal{S}_{\lambda}(x) = \operatorname{sign}(x) \cdot \max(|x| - \lambda, 0)$$

```
function out = shrinkage(k, a)
    out = zeros(size(a));
    case1 = a > k;
    out(case1) = a(case1) - k;
    case2 = a < -k;
    out(case2) = a(case2) + k;
end
```

Main Function

This function will return α, where x is the input signal

```
function alpha=sig_seperate(x)
rho=1;
len=2*length(x);
z0=randi([-3 3],len,1);
lambda0=randi([1 3],len,1);
max_iter=1000;
tol=1e-5;
A=ocd(x);
for i=1:max_iter
    alpha=shrinkage(1/rho,z0-(1/rho)*lambda0);
    z1=(eye(len)-pinv(A)*A)*(alpha + (1/rho)*lambda0) + pinv(A)*x;
    lambda1=lambda0 + rho*(alpha - z1);
    if norm(z1-z0)<=tol
        break
    end
    z0=z1;
```

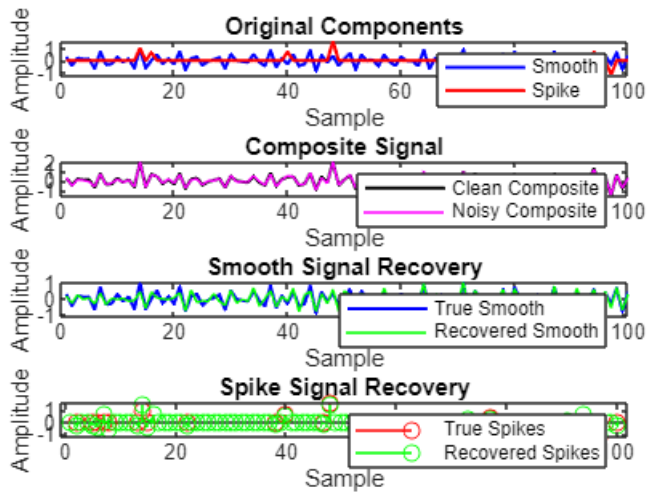
```
    lambda0=lambda1;
end
end
```

Experimentation

20 dB case with k_1=5, k_2=10, N=100

```
clc; clear; close all;
N = 100;
k1 = 5;
k2 = 10;
snr_db = 20;
v_dct = zeros(N,1);
idx1 = randperm(N, k1);
v_dct(idx1) = randn(k1,1);
v = idct(v_dct);
w = zeros(N,1);
idx2 = randperm(N, k2);
w(idx2) = randn(k2,1);
x = v + w;
x_noisy = awgn(x, snr_db, 'measured');
alpha = sig_seperate(x_noisy);
len = length(alpha)/2;
alpha_smooth = alpha(1:len);
alpha_spike  = alpha(len+1:end);
v_rec = idct(alpha_smooth);
w_rec = alpha_spike;
x_rec = v_rec + w_rec;
figure('Name','Spike + Smooth Separation (20 dB)','NumberTitle','off');
subplot(4,1,1);
plot(v,'b','LineWidth',1.2); hold on;
plot(w,'r','LineWidth',1.2);
legend('Smooth','Spike');
title('Original Components');
xlabel('Sample'); ylabel('Amplitude');

subplot(4,1,2);
plot(x,'k','LineWidth',1.2); hold on;
plot(x_noisy,'m');
legend('Clean Composite','Noisy Composite');
title('Composite Signal');
xlabel('Sample'); ylabel('Amplitude');

subplot(4,1,3);
plot(v,'b','LineWidth',1.2); hold on;
plot(v_rec,'g');
legend('True Smooth','Recovered Smooth');
```

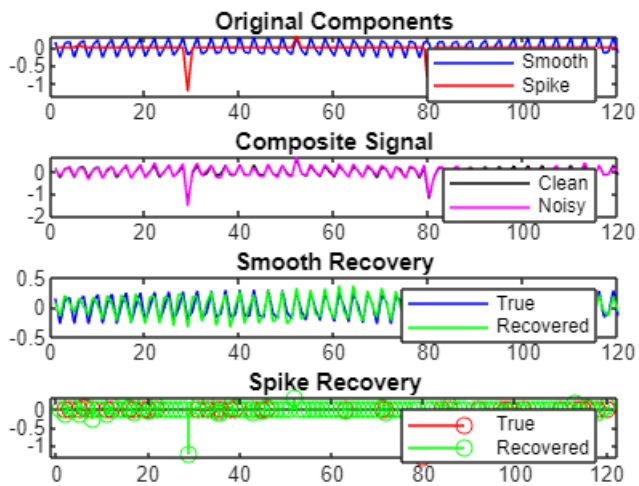
```
title('Smooth Signal Recovery');
xlabel('Sample'); ylabel('Amplitude');

subplot(4,1,4);
stem(w,'r'); hold on;
stem(w_rec,'g');
legend('True Spikes','Recovered Spikes');
title('Spike Signal Recovery');
xlabel('Sample'); ylabel('Amplitude');
hold off;
```



12 dB case with k_1=3, k_2=5, N=120

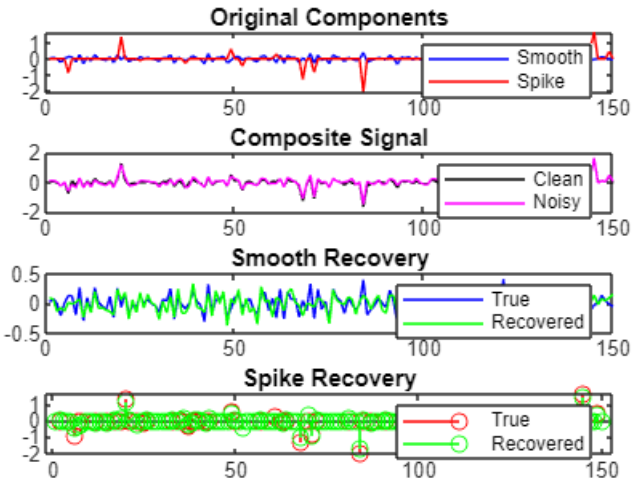
```
clc; clear; close all;
N = 120;
k1 = 3;
k2 = 5;
snr_db = 12;
v_dct = zeros(N,1);
v_dct(randperm(N,k1)) = randn(k1,1);
v = idct(v_dct);
w = zeros(N,1);
w(randperm(N,k2)) = randn(k2,1);
x = v + w;
x_noisy = awgn(x, snr_db, 'measured');
alpha = sig_seperate(x_noisy);
len = length(alpha)/2;
v_rec = idct(alpha(1:len));
w_rec = alpha(len+1:end);
figure('Name','Separation (12 dB)','NumberTitle','off');
subplot(4,1,1); plot(v,'b'); hold on; plot(w,'r'); legend('Smooth','Spike');
title('Original Components');
subplot(4,1,2); plot(x,'k'); hold on; plot(x_noisy,'m'); legend('Clean','Noisy');
title('Composite Signal');
```

```
subplot(4,1,3); plot(v,'b'); hold on; plot(v_rec,'g'); legend('True','Recovered');
title('Smooth Recovery');
subplot(4,1,4); stem(w,'r'); hold on; stem(w_rec,'g'); legend('True','Recovered');
title('Spike Recovery');
```



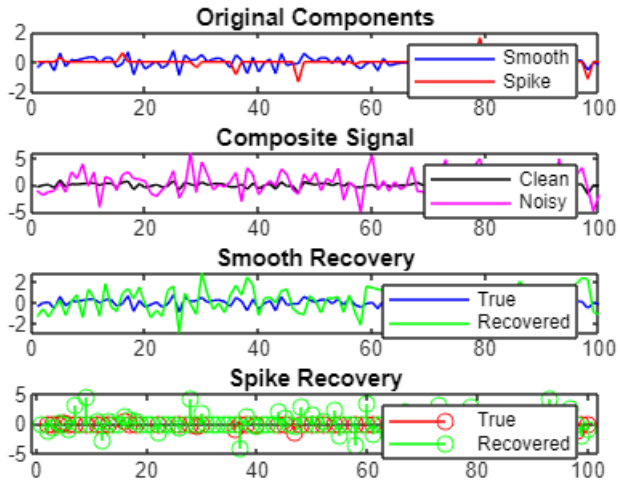
10 dB case with k_1=8, k_2=20, N=150

```
clc; clear; close all;
N = 150; k1 = 8; k2 = 20; snr_db = 10;

v_dct = zeros(N,1); v_dct(randperm(N,k1)) = randn(k1,1);
v = idct(v_dct);

w = zeros(N,1); w(randperm(N,k2)) = randn(k2,1);

x = v + w; x_noisy = awgn(x, snr_db, 'measured');

alpha = sig_seperate(x_noisy);
len = length(alpha)/2; v_rec = idct(alpha(1:len)); w_rec = alpha(len+1:end);

figure('Name','Separation (10 dB)','NumberTitle','off');
subplot(4,1,1); plot(v,'b'); hold on; plot(w,'r'); legend('Smooth','Spike');
title('Original Components');
subplot(4,1,2); plot(x,'k'); hold on; plot(x_noisy,'m'); legend('Clean','Noisy');
title('Composite Signal');
subplot(4,1,3); plot(v,'b'); hold on; plot(v_rec,'g'); legend('True','Recovered');
title('Smooth Recovery');
subplot(4,1,4); stem(w,'r'); hold on; stem(w_rec,'g'); legend('True','Recovered');
title('Spike Recovery');
```



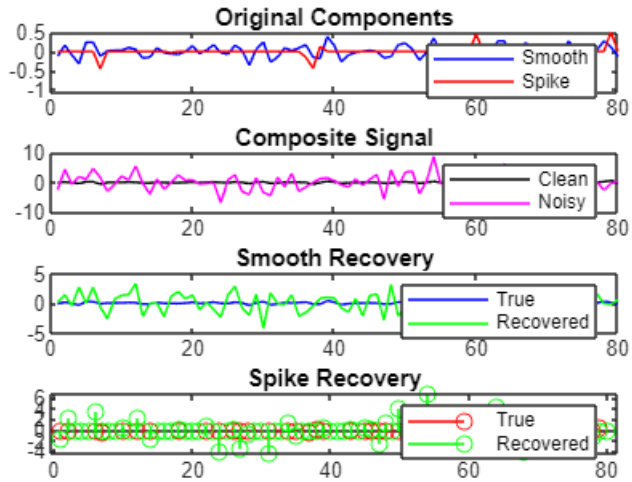
-12 dB case with k_1=15, k_2=10, N=100

```
clc; clear; close all;
N = 100; k1 = 15; k2 = 10; snr_db = -12;

v_dct = zeros(N,1); v_dct(randperm(N,k1)) = randn(k1,1);
v = idct(v_dct);

w = zeros(N,1); w(randperm(N,k2)) = randn(k2,1);

x = v + w; x_noisy = awgn(x, snr_db, 'measured');

alpha = sig_seperate(x_noisy);
len = length(alpha)/2; v_rec = idct(alpha(1:len)); w_rec = alpha(len+1:end);

figure('Name','Separation (-12 dB)','NumberTitle','off');
subplot(4,1,1); plot(v,'b'); hold on; plot(w,'r'); legend('Smooth','Spike');
title('Original Components');
subplot(4,1,2); plot(x,'k'); hold on; plot(x_noisy,'m'); legend('Clean','Noisy');
title('Composite Signal');
subplot(4,1,3); plot(v,'b'); hold on; plot(v_rec,'g'); legend('True','Recovered');
title('Smooth Recovery');
subplot(4,1,4); stem(w,'r'); hold on; stem(w_rec,'g'); legend('True','Recovered');
title('Spike Recovery');
```



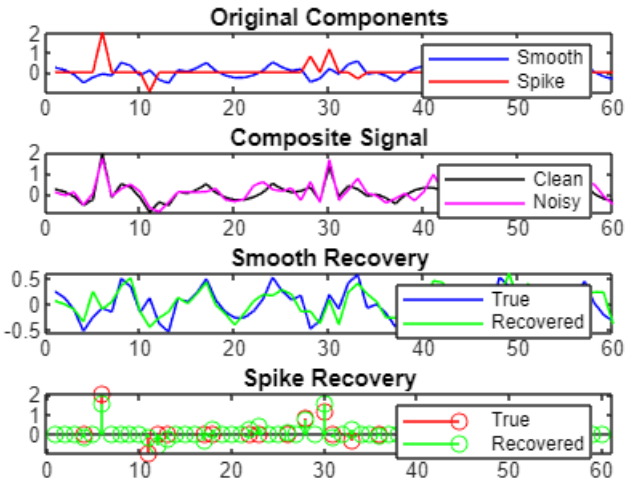
-20 dB case with k_1=8, k_2=8, N=80

```
clc; clear; close all;
N = 80; k1 = 8; k2 = 8; snr_db = -20;

v_dct = zeros(N,1); v_dct(randperm(N,k1)) = randn(k1,1);
v = idct(v_dct);

w = zeros(N,1); w(randperm(N,k2)) = randn(k2,1);

x = v + w; x_noisy = awgn(x, snr_db, 'measured');

alpha = sig_seperate(x_noisy);
len = length(alpha)/2; v_rec = idct(alpha(1:len)); w_rec = alpha(len+1:end);

figure('Name','Separation (-20 dB)','NumberTitle','off');
subplot(4,1,1); plot(v,'b'); hold on; plot(w,'r'); legend('Smooth','Spike');
title('Original Components');
subplot(4,1,2); plot(x,'k'); hold on; plot(x_noisy,'m'); legend('Clean','Noisy');
title('Composite Signal');
subplot(4,1,3); plot(v,'b'); hold on; plot(v_rec,'g'); legend('True','Recovered');
title('Smooth Recovery');
subplot(4,1,4); stem(w,'r'); hold on; stem(w_rec,'g'); legend('True','Recovered');
title('Spike Recovery');
```



6 dB case with k_1=4, k_2=6, N=60

```
clc; clear; close all;
N = 60; k1 = 4; k2 = 6; snr_db = 6;

v_dct = zeros(N,1); v_dct(randperm(N,k1)) = randn(k1,1);
v = idct(v_dct);

w = zeros(N,1); w(randperm(N,k2)) = randn(k2,1);

x = v + w; x_noisy = awgn(x, snr_db, 'measured');

alpha = sig_seperate(x_noisy);
len = length(alpha)/2; v_rec = idct(alpha(1:len)); w_rec = alpha(len+1:end);

figure('Name','Separation (6 dB)','NumberTitle','off');
subplot(4,1,1); plot(v,'b'); hold on; plot(w,'r'); legend('Smooth','Spike');
title('Original Components');
subplot(4,1,2); plot(x,'k'); hold on; plot(x_noisy,'m'); legend('Clean','Noisy');
title('Composite Signal');
subplot(4,1,3); plot(v,'b'); hold on; plot(v_rec,'g'); legend('True','Recovered');
title('Smooth Recovery');
subplot(4,1,4); stem(w,'r'); hold on; stem(w_rec,'g'); legend('True','Recovered');
title('Spike Recovery');
```



Future Work

Changing project due to simple work in this paper